

Course Code: ECE3006	Course Title HDL Verification and Methodology	TPC	3	2	4
Version No.	1.1				
Course Pre-requisites/ Co-requisites/ anti-requisites (if any). Otherwise, please indicate as 'None'	ECE1003				
Objectives:	The objectives of this course are: 1. To equip student with various concepts of verification in digital VLSI. 2. To equip student ability to work on various verification projects using system Verilog. 3. To enable the students to verify and design the digital circuit with the aid of Computer Aided Engineering tools which involves in programming with the help of Verilog HDL. 4. To enable the students to simulate, synthesize, and program their designs on a development board.				
CO's Mapping with PO's and PEO's					
Course Outcomes	Course Outcome Statement	PO's / PEO's			
CO1	Acquire knowledge on verification and application for VLSI designs.	PO1, PO3, PO5, PEO1			
CO2	Acquire knowledge on verification methodologies and different types of simulators.	PO1, PO2, PEO1, PEO2			
CO3	Design a solution to obtain 100% code coverage & functional coverage by determining the set of input constraints and assertions in test benches.	PO3, PO4, PO5, PEO1, PEO2, PEO3			
CO4	To understand the basic instructions and syntax of System Verilog.	PO1, PO2, PEO1, PEO2			
CO5	To simulate the test bench architecture using system Verilog.	PO3, PO4, PO5, PEO1, PEO2			
CO6	To get acquainted with the universal verification methodology.	PO1, PO2, PO3, PO4, PO5, PEO1, PEO2			
CO7	To simulate, verify and program different VLSI designs.	PO1, PO2, PO3, PO4, PO5, PEO1, PEO2			
		TOTAL HOURS OF INSTRUCTIONS: 45			
Module No. 1	Importance of Verification and different approaches	6 Hours			

Concepts of verification, importance of verification, convergence model, Formal verification, Equivalence checking, Model checking, Functional verification, Functional verification approaches, testing versus verification: scan based testing, design for verification. Verification reuse. The cost of verification		
Module No. 2	Code & Functional Coverage and verification plan:	6 Hours
Statement coverage, path coverage, expression coverage, FSM coverage, what does 100%coverage mean? Item Coverage, cross coverage, Transition coverage, what does 100% functional mean? Assertions, Issue tracking & Metrics. The role of verification plan: specifying the verification plan, defining the first success. ASIC verification categories- block level verification, system level verification, board level verification, verifying strategies, Reusable components verification.		
Module No. 3	Introduction to System Verilog	13 Hours
Data Types, Procedural Statements and Routines. Connecting the Test bench and Design, Basic OOP, Randomization, functional coverage using System Verilog.		
Module No. 4	Verification and Methodologies using System Verilog	6 Hours
Advanced OOP and Test bench Guidelines, Advanced Interfaces, A Complete System Verilog Test bench, Introduction to Verification Methodologies, System Verilog Assertions.		
Module No. 5	Universal Verification Methodology:	14 Hours
Introduction to Universal Verification Methodology, Overview of UVM Base Classes and simulation phases in UVM and UVM macros, UVM Transaction ,TLM Port		
Text Books 1. Janick Bergeron, —Writing test benches: functional verification of HDL models, 2nd edition, Kluwer Academic Publishers, 2012		
References 1. SystemVerilog for Verification by Chris Spear ,3rd edition Springer Publisher 2. Hardware Verification with System Verilog by Mintz, Mike, Ekendahl, Robert 3. Writing Testbenches using SystemVerilog by Bergeron, Janick 4. The UVM Primer by Ray Salemi 5. The Verification Methodology Cookbook - online textbook 6. For Formal Verification - "Formal Verification - An Essential Toolkit for Modern VLSI Design"		
List of Laboratory Experiments: Verification of the following circuits using system Verilog to obtain 100% code coverage & functional coverage by determining the set of input constraints and assertions in test benches. 1. Decoder, Encoders, Mux and Demux 2. Adders, Subtractors and ALU 3. Flip Flop and Latches 4. Synchronous and asynchronous Counters 5. Single port and dual port synchronous and asynchronous Memories 6. Parity and CRC 7. Verification of FIFO 8. Decoder, Encoders, Mux and Demux using UVM 9. Verification of counter using UVM 10. Verification of Switch using UVM		

Mode of Evaluation	Continuous Assess Continuous Assessment (CATs and FAT – 75%; Quizzes, Assignments, Projects, etc. – 25%).
Recommended by the Board of Studies on	05.02.2021
Date of Approval by the Academic Council	

PO/PSO CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2	PSO3
C01	3	3											3		
C02	3	3											3		3
C03			3	3	2								3		
C04	3	3											3		
C05				3	2								3	3	
C06	3	3		3	2								3	3	
C07	3	3		3	2								3	3	3